

Conservative systems & the phase plane

Ex. $m\ddot{x} = F(x)$, $x \in \mathbb{R}$

let $V(x) = -\int F(x) dx$

$\Rightarrow m\ddot{x} = -V'(x)$

conserved quantity:

$$E = \frac{1}{2} m \dot{x}^2 + V(x)$$

check: $\frac{dE}{dt} = m\dot{x}\ddot{x} + V'(x)\dot{x} = \dot{x} [m\ddot{x} + \underbrace{V'(x)}_{-V'(x)}] = 0$

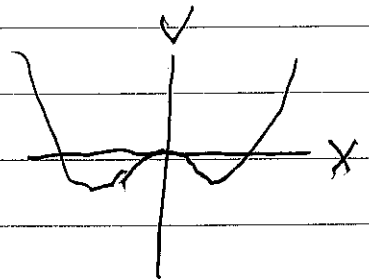
consider $V(x) = -\frac{1}{2}\alpha x^2 + \frac{1}{4}\beta x^4$, $\alpha, \beta > 0$

$$\ddot{x} = \frac{\alpha}{m} x - \frac{\beta}{m} x^3 = \frac{\alpha}{m} \left[x - \frac{\beta}{\alpha} x^3 \right]$$

let $t = \sqrt{\frac{m}{\alpha}} \tau$, $x = \sqrt{\frac{\alpha}{\beta}} X$

$$\frac{d^2 X}{d\tau^2} = \sqrt{\frac{\beta}{\alpha}} \frac{d^2 x}{dt^2} \left(\frac{dt}{d\tau} \right)^2 = \sqrt{\frac{\beta}{\alpha}} \frac{m}{\alpha} \frac{d^2 x}{dt^2} = \sqrt{\frac{\beta}{\alpha}} \left[\sqrt{\frac{\alpha}{\beta}} X - \frac{\beta}{\alpha} \left(\frac{\alpha}{\beta} \right)^{3/2} X^3 \right]$$

$$\frac{d^2 X}{d\tau^2} = X - X^3, \quad V = -\frac{1}{2} X + \frac{1}{4} X^4$$



Re-write as 2 first-order eqns.

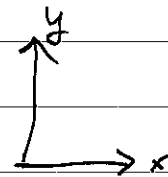
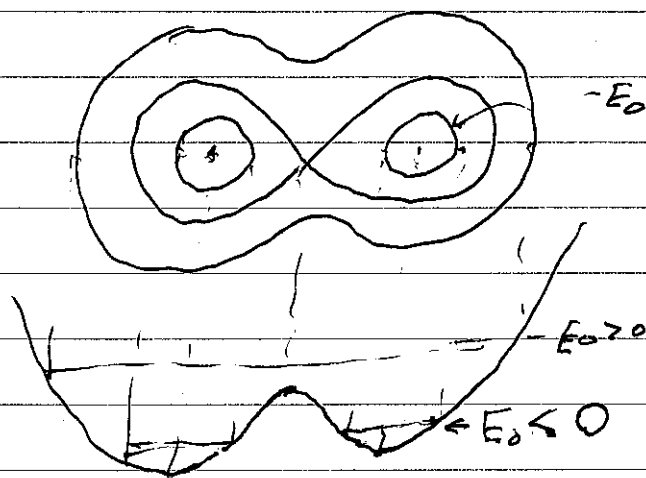
$$\dot{X} = Y$$

$$\dot{Y} = X - X^3$$

$$\frac{1}{2} Y(t)^2 - \frac{1}{2} X(t)^2 + \frac{1}{4} X(t)^4 = \frac{1}{2} Y_0^2 - \frac{1}{2} X_0^2 + \frac{1}{4} X_0^4$$

Wolfram α | pha (Mathematica)

ContourPlot [$\frac{1}{2} y^2 - \frac{1}{2} x^2 + \frac{1}{4} x^4$, {x, -2, 2}, {y, -3, 3}]



$$(0, 0) \Rightarrow E_0 = 0$$

$$(\pm 1, 0) \Rightarrow E_0 = -\frac{1}{4}$$

$(0, 0)$ is a saddle (unstable)
 $(\pm 1, 0)$ are centers (Lyapunov stable)

add damping $\dot{X} = -\epsilon \dot{X} + X - X^3$ $0 < \epsilon \ll 1$

$$\dot{X} = Y$$

$$\dot{Y} = -\epsilon Y + X - X^3$$

same equilibria

$$(0, 0), (\pm 1, 0)$$

$$DF = \begin{bmatrix} 0 & 1 \\ 1-3x^2 & -\epsilon \end{bmatrix}$$

$$DF(0,0) = \begin{bmatrix} 0 & 1 \\ 1 & -\epsilon \end{bmatrix}$$

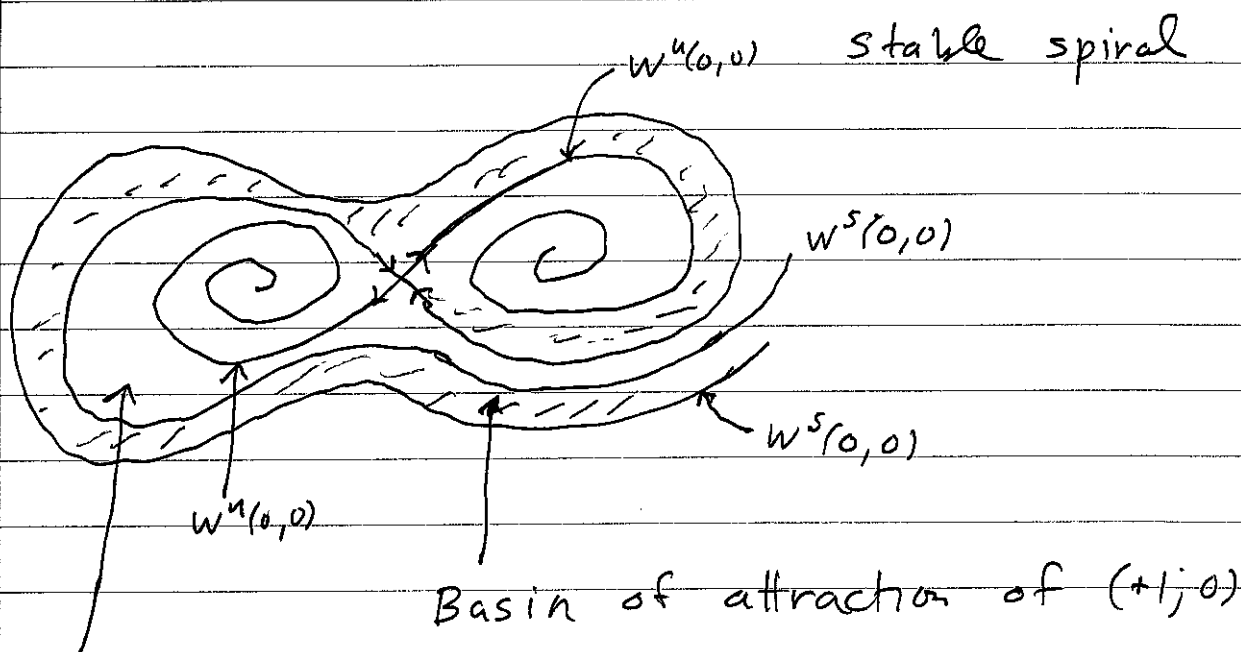
$$\left. \begin{array}{l} \tau = -\epsilon \\ \Delta = -1 \end{array} \right\} \text{ saddle}$$

$$DF(\pm 1, 0) = \begin{bmatrix} 0 & 1 \\ -2 & -\epsilon \end{bmatrix}$$

$$\left. \begin{array}{l} \tau = -\epsilon \\ \Delta = 2 \end{array} \right\} \text{ stable}$$

$$\tau^2 - 4\Delta = \epsilon^2 - 8 < 0$$

for $\epsilon \ll 1$



$w^s(0,0)$ forms the basin boundaries for stable fixed-pts $(\pm 1, 0)$

Properties of stable/unstable manifolds.
 $W^s(x^*)$, $W^u(x^*)$ of fixed pt x^*

- if $x_0 \in W^s(x^*)$, then $\lim_{t \rightarrow \infty} x(t) = x^*$
- if $x_0 \in W^u(x^*)$, then $\lim_{t \rightarrow -\infty} x(t) = x^*$
- dynamically invariant - if $x_0 \in W^s(x^*)$ then $x(t) \in W^s(x^*) \quad \forall t > 0$
- Tangent to E^s/E^u at x^*
- Can be represented as a graph over E^s/E^u in neighborhood of x^*

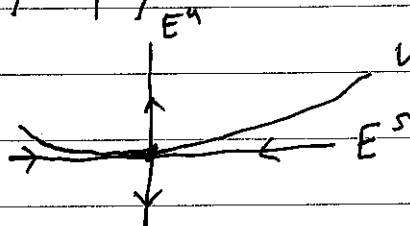
Ex. $\dot{x} = -\lambda x + Ax^2 + Bxy$
 $\dot{y} = \mu y + Bx^2$

$\lambda, \mu > 0$

linearization:

$\dot{x} = -\lambda x \leftarrow \text{stable}$

$\dot{y} = \mu y \leftarrow \text{unstable}$



$W^s = \{(x, h(x)), |x| \leq 1\}$

$y = h(x)$ is tangent
 to x -axis at $(0,0)$
 $h(0) = 0, h'(0) = 0$

Invariant: if $(x_0, y_0) = (x_0, h(x_0))$ so
initial condition in $W^s(x^*)$ then

$$(x(t), y(t)) = (x(t), h(x(t)))$$

$$y(t) = h(x(t))$$

$$\frac{dy}{dt} = \frac{dh}{dx} \frac{dx}{dt}$$

$$\mu y + Bx^2$$

$$y = h(x)$$

$$-\lambda x + Ax^2 + Bxy$$

$$y = h(x)$$

$$\mu h(x) + Bx^2 = h'(x) [-\lambda x + Ax^2 + Bxh(x)]$$

diff. eqn. for $h(x)$ where $h(0) = h'(0) = 0$

This is valid for $|x| \ll 1$, so try a power series soln. for it

$$h = \overset{0}{x} + \overset{0}{\beta x} + \gamma x^2 + \delta x^3 + \dots$$

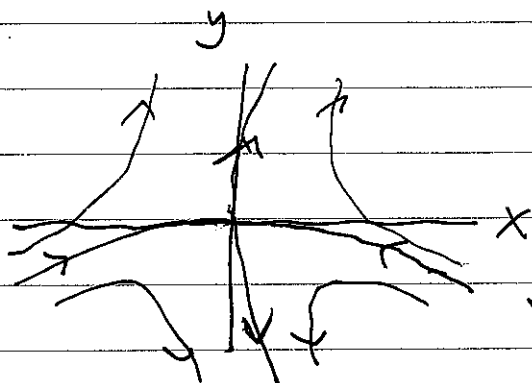
because stable manifold is tangent to E^s at $(h(0) = h'(0) = 0)$

$$\mu (\gamma x^2 + \delta x^3 + O(4)) + Bx^2 = [2\gamma x + 3\delta x^2 + O(3)]$$

$$[-\lambda x + Ax^2 + Bx(\gamma x^2) + O(4)]$$

$$O(x^2): \mu \gamma + B = -2\gamma \lambda \Rightarrow B = -(2\lambda + \mu)\gamma \Rightarrow \gamma = \frac{-B}{(2\lambda + \mu)}$$

$$h = -\frac{B}{(\mu+2\lambda)} x^2 + O(x^3) \quad \left. \vphantom{h} \right\} \begin{array}{l} \text{Note:} \\ \text{this is not} \\ \text{the } y\text{-nullcline,} \\ \text{which is } y = -\frac{B}{\mu} x^2 \end{array}$$



← case $B > 0$

$$W^u(0,0) = \{ (g(y), y) : |y| \leq 1 \}$$

$x = g(y)$

$$g(0) = g'(0) = 0$$

$x = g(y)$ is also dynamically invariant ---

~~left over from fast time~~